PHYC10009 Physics Foundations

Capacitance, dielectrics, electric current
Week 8 Lecture 1

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Today

- Capacitance and capacitors
- Dielectrics
- Stored electric energy
- Electric current

Similarities

Recall four similar equations

- Coulomb force $F = k\dfrac{Qq}{r^2}$
- Electric field $E = k\dfrac{Q}{r^2}$ (around a point charge $+Q$)
- Potential energy $\mathrm{PE} = k\dfrac{Qq}{r}$ (for a charge q in field of a point charge $+Q$)
- Voltage $V = k\dfrac{Q}{r}$ (around a point charge $+Q$)

Field and voltage associated with just one charge

Force and energy associated with two charges

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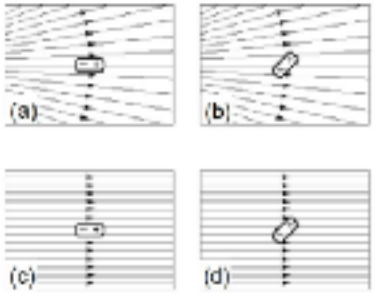
Force on Electric Dipole

An electrically neutral dipole is placed in an external field. In which situation(s) is the net force on the dipole zero?



1(a)

2 (c)

3 (b) and (c)

4 (a) and (c)

5 (c) and (d)

6 some other combination

7 none of the above

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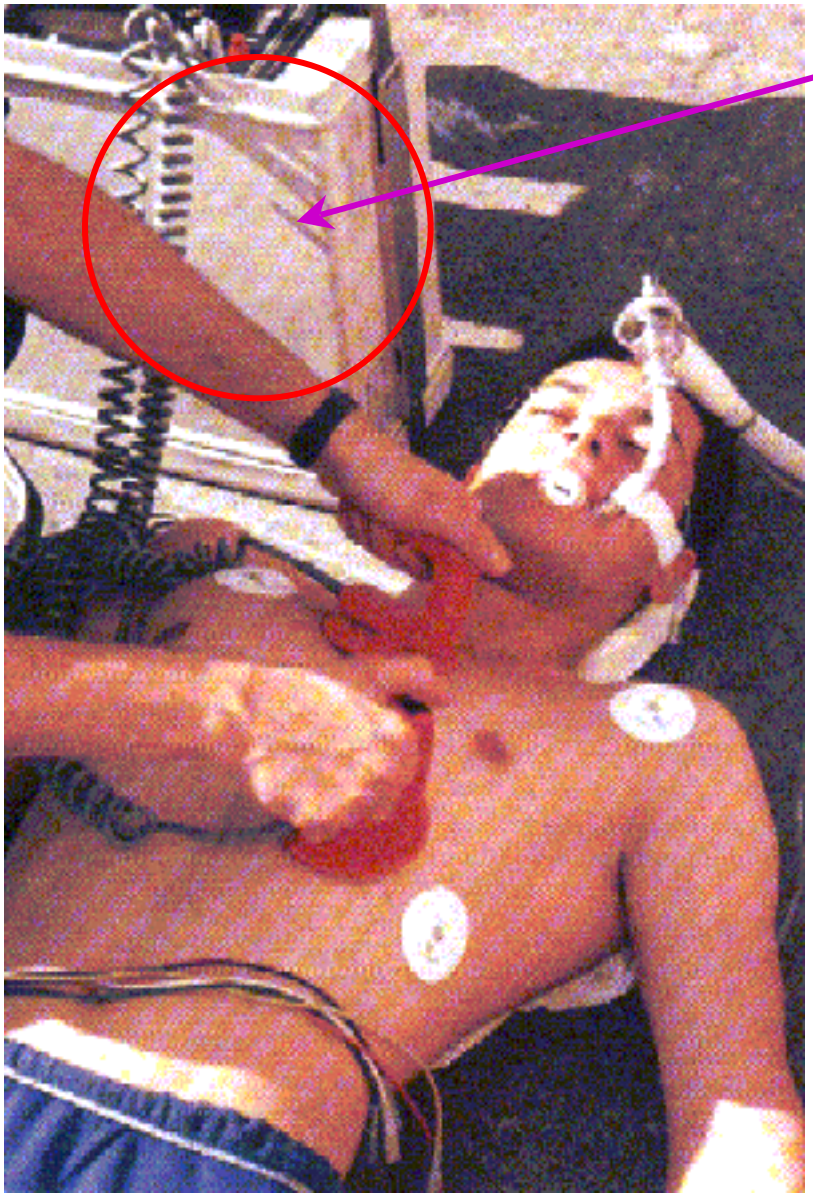
Capacitance and capacitors

Capacitor example



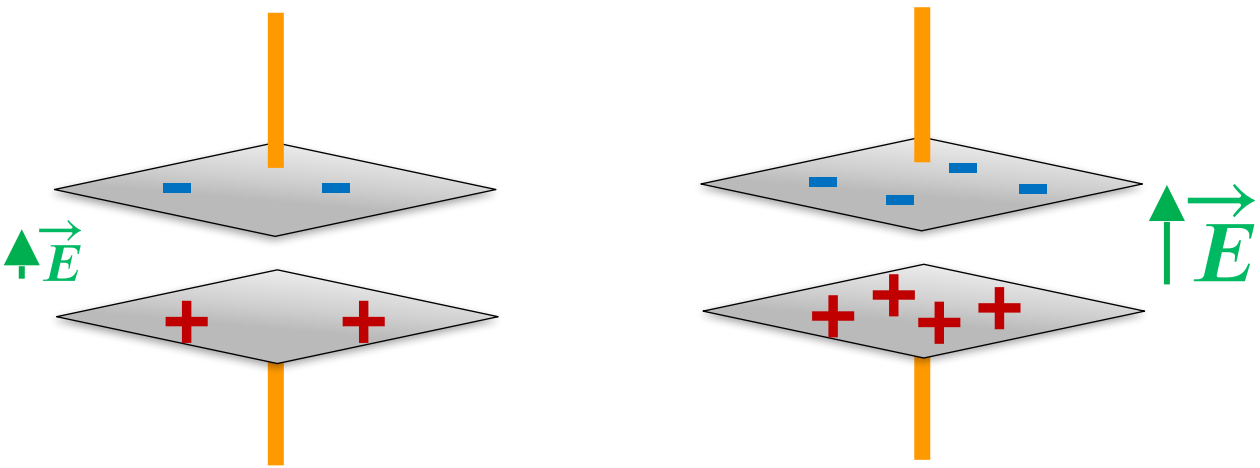
- There's one in here…
- The heart in this accident victim has either stopped or is beating erratically
- 20 Amperes of electric current must be sent through the chest cavity
- This will shock the heart back into normal activity
- This shock amounts to the transfer of 200 Joules of electrical energy in only 2 milliseconds
- 100 kW of electrical power!!
- Where does this come from?

17.7 Capacitance

- Capacitance is critical to nerve and muscle function
- *Heart defibrillator:* employs a large capacitor to deliver an electrical pulse

Capacitor = separation of opposite charge, thus potential energy

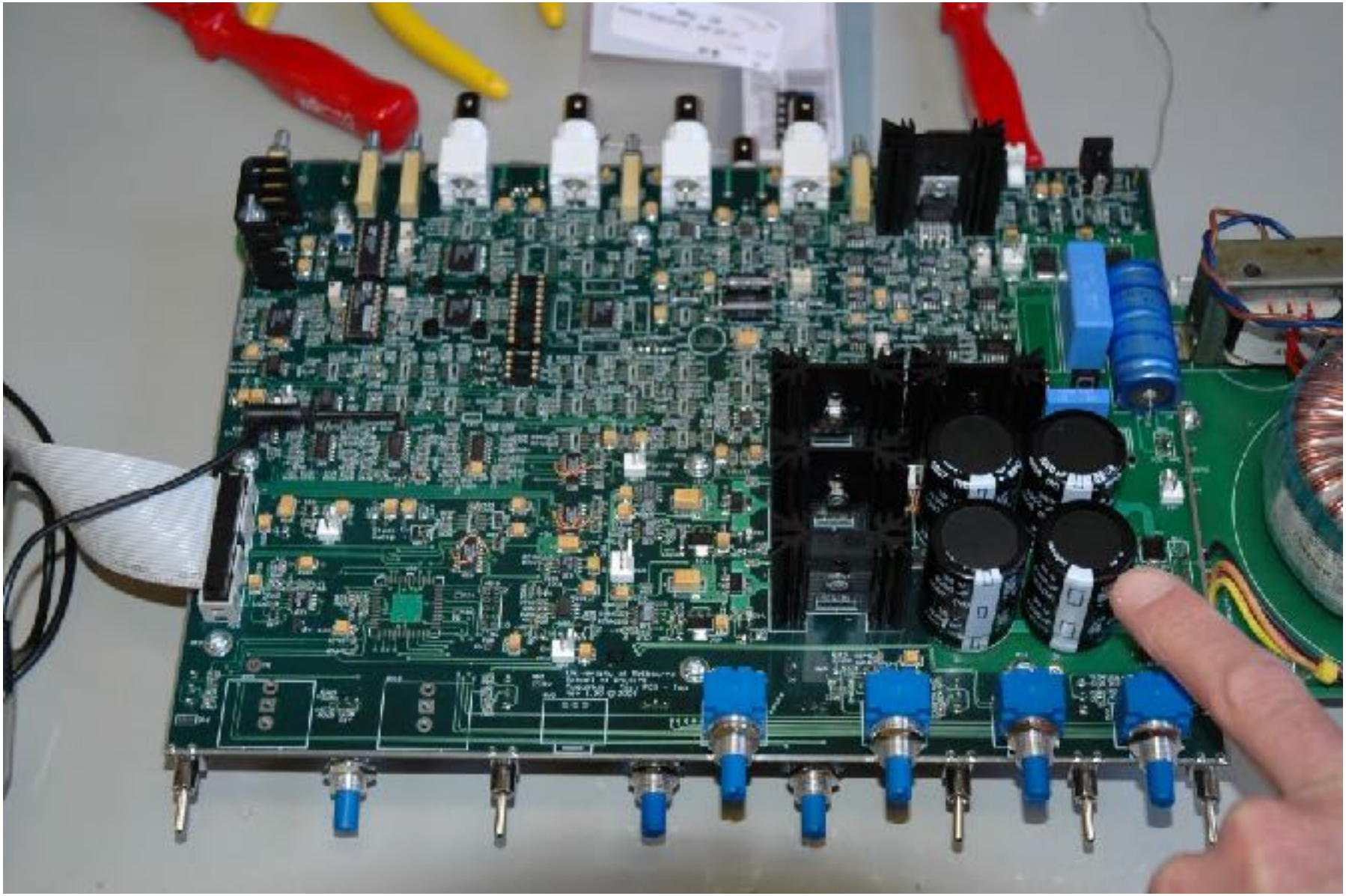
- i.e. push charge $+Q$ to a conductor, charge $-Q$ to another conductor in close proximity



Small field
Small V

Larger field
Larger V

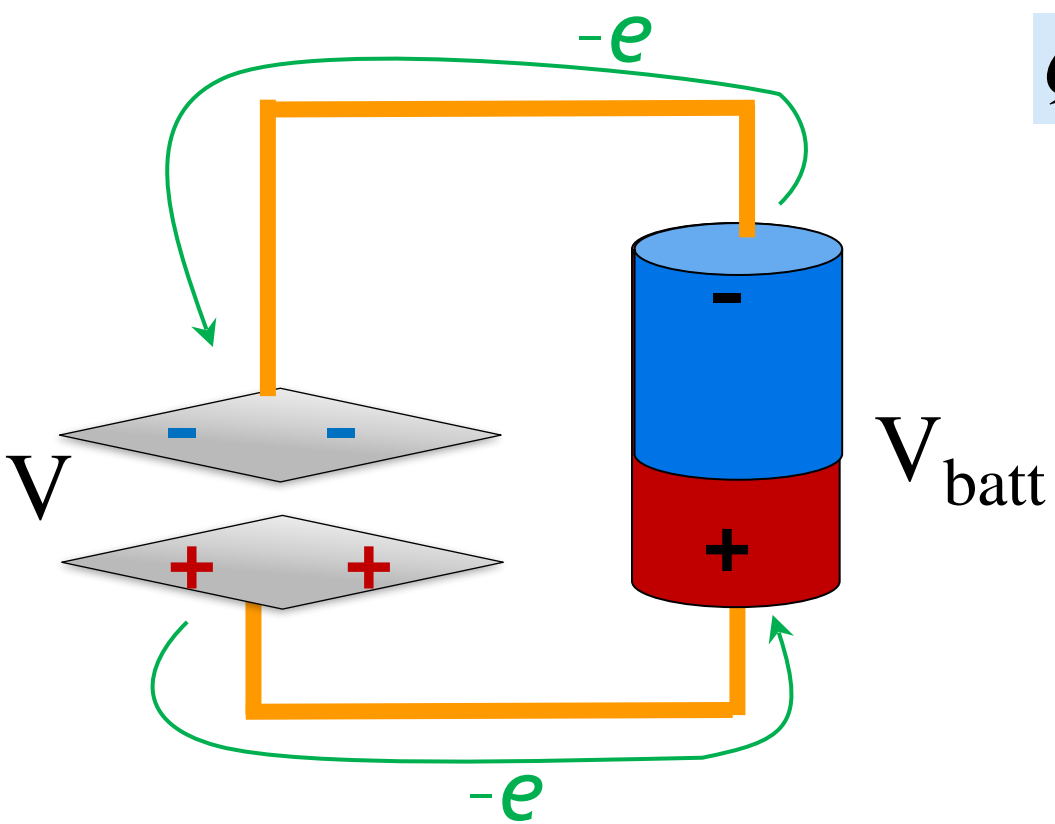
About 700 capacitors...



From 10pF to 8.2mF

Capacitance

- As the charge Q increases, so does the electric field: $E = \dfrac{Q}{\varepsilon_0 A}$
- As the electric field increases so does the voltage… until $V = V_{\text{batt}}$

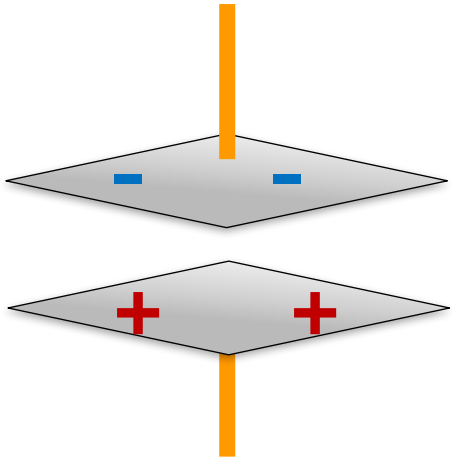


$$Q = CV$$

$$C = \frac{\varepsilon_0 A}{d}$$

(parallel plates)

Disconnected:
charge remains



$$V = V_{\text{batt}}$$

D El-4 The Capacitor (showing ΔV after removing power)

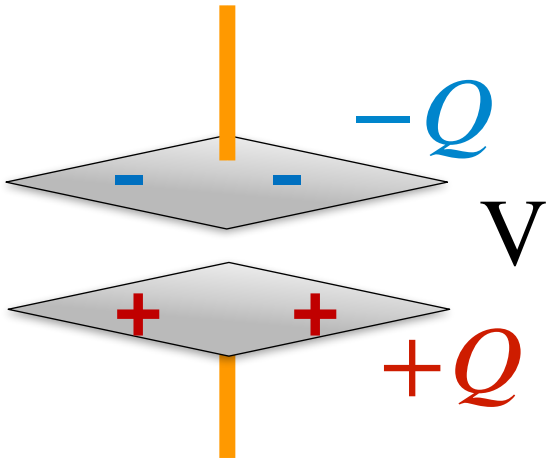
Capacitance

$$Q = CV$$

Q = charge, Coulombs

V = volts

C = capacitance = **Farads**



Capacitance is a measure of capacity to store energy.

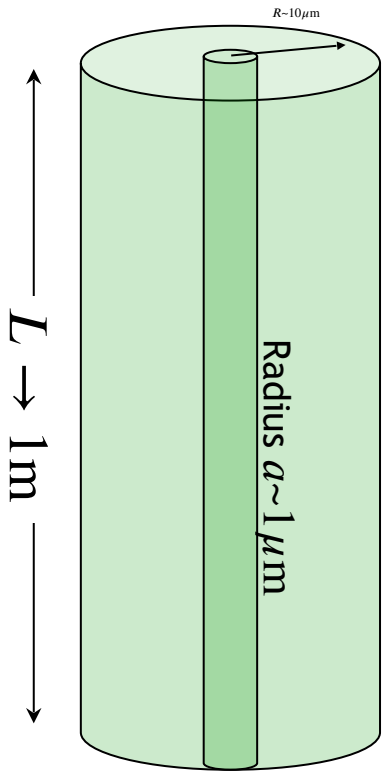
Here we have stored energy in electric field between the plates.

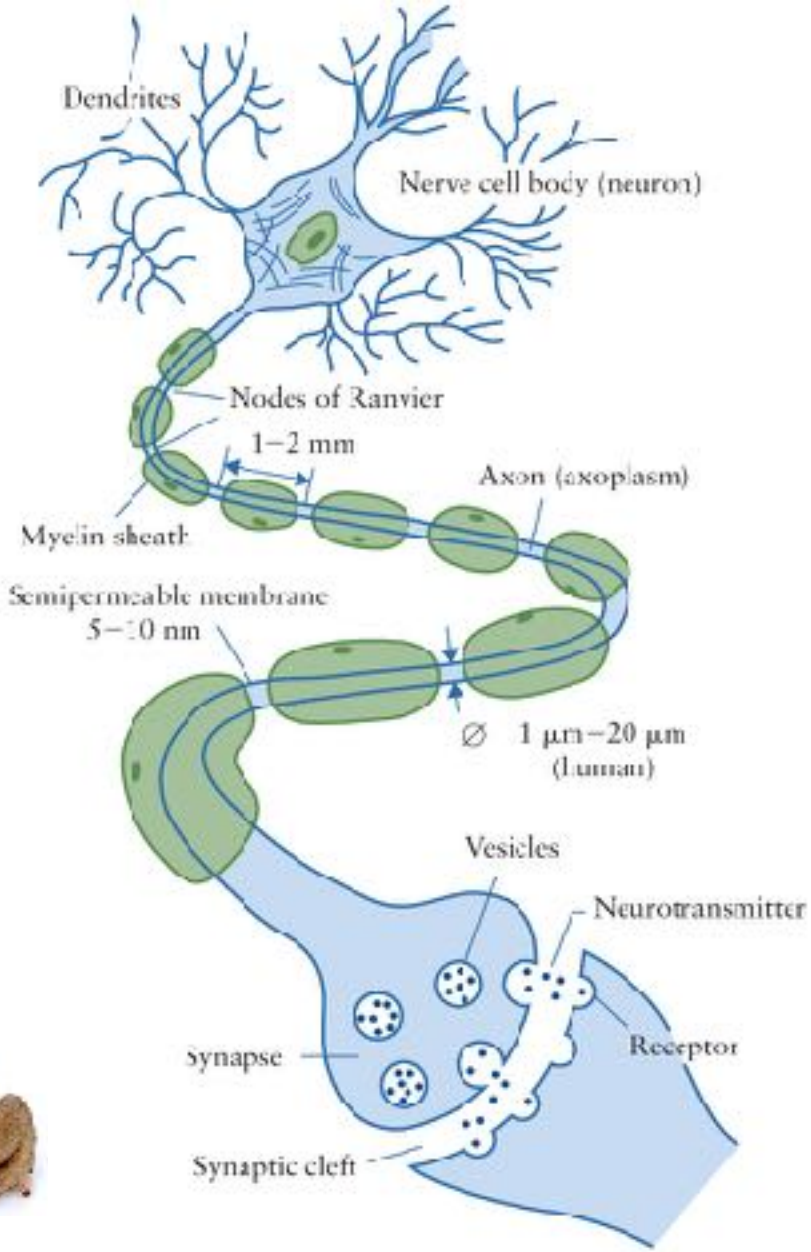
Demo E field, capacitor

Demo capacitor discharge

Really small capacitors

- Nerve axons are capacitors, up to 1 m long, ~ 10μm in diameter
- Capacitance ~ tens of nF





Capacitor: example

Consider a 1 F capacitor with plate spacing of 1 mm

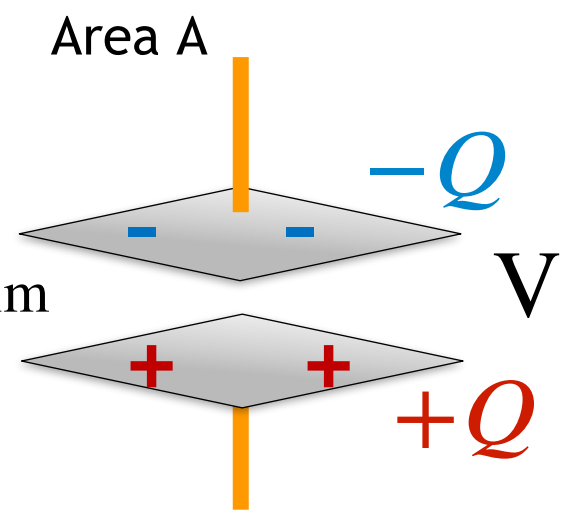
(a) What is the surface area of the plates?

(b) How much charge is on the plates if this capacitor is attached to a 1.5V battery?

Area: $C = \dfrac{\varepsilon_0 A}{d}$

$$A = \frac{Cd}{\varepsilon_0} = \frac{0.001}{8.85 \times 10^{-12}} = 1.13 \times 10^8 \mathrm{m}^2 \sim 10 \times 10 \text{ km}$$

Area is ***very*** large!



Charge: $Q = CV = 1.5$ C (about 10^{19} electrons)

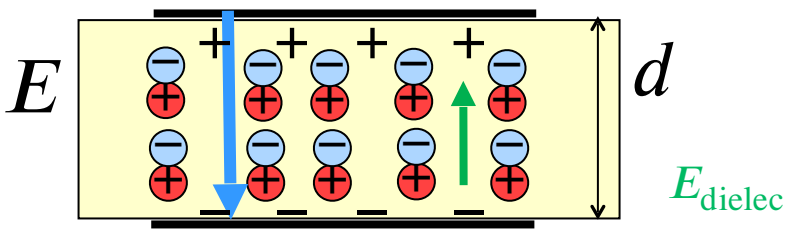
17.8 Dielectric Materials

Huge areas?

Trick: reduce the electric field for a given Q: $Q = CV = CEd$ so $C = \dfrac{Q}{Ed}$

Dielectric materials

Electrical insulators reduce electric field by factor κ = dielectric constant



$C = \dfrac{Q}{Ed}$ but electric field $\rightarrow$ $E - E_{\text{dielec}}$

$$C' = \kappa C$$

κ is called the dielectric constant

κ always ≥ 1

EL-4 Demo dielectric

Dielectric Materials

Some Dielectric Materials

Material	Dielectric Constant κ
Vacuum	1 (exactly)
Air	1.00054
Teflon	2.0
Paper	3.0
Pyrex glass	4.8
Cell membrane	7 to 9
Ethanol	24
Tantalum pentoxide (tantalum caps)	26
Water	80
Strontium titanate	310
Barium titanate	Up to 20,000

Dielectric materials: example

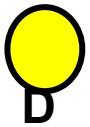
A parallel-plate capacitor is charged using a 100 V battery, then the battery is removed. If a dielectric slab is slid between the plates, the capacitor voltage drops to 20 V.

What is the dielectric constant of the dielectric?

Electric field strength is $E = V/d$

Distance d doesn't change, so: $$\frac{E_1}{E_2} = \frac{V_1}{V_2} = \frac{100}{20} = 5$$

We can write $E_1 = \kappa E_2$ thus $\kappa = \dfrac{E_1}{E_2} = 5$



El-4 The capacitor with pyrex glass showing effect of dielectric

17.9 Energy and Capacitors

Energy storage: potential energy of displaced charge

As we push more charge onto the capacitor, voltage difference increases; more and more difficult to add charge.

$$\mathrm{PE} = \int_0^Q V(q)\,dq = \int_0^Q \frac{q}{C}\,dq = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}\frac{(CV)^2}{C} = \frac{1}{2}CV^2$$

$$\mathrm{PE} = \frac{1}{2}CV^2$$

PE = potential energy
Q = total charge
C = capacitance
V = voltage difference

Defibrillators store up to 360 J

D Ed-2 Energy stored in a capacitor

Demo: Energy in Capacitors.

Example: 2008 exam

(c) The capacitor in a defibrillator is initially charged to 5000V at which point it stores 250 Joules of energy.

(i) What is the capacitance of the device?

$$\text{PE} = \frac{1}{2}CV^2 = 250 \text{ J}$$

$$\Rightarrow C = \frac{2 \times 250}{5000^2} = 20\ \mu F$$

Example: Energy in capacitors

The electrostatic energy in a charged capacitor is 300 J.

The capacitor is discharged in a duration of 2ms.

(a) If capacitance is 40μF, what is the voltage difference on the cap?
(b) What is the average power delivered?

(a) so

(b) Power is energy over time:

Example: Energy in capacitors

The electrostatic energy in a charged capacitor is 300 J.

The capacitor is discharged in a duration of 2ms.

(a) If capacitance is 40μF, what is the voltage difference on the cap?

(b) What is the average power delivered?

(a) $\mathrm{PE} = \frac{1}{2}CV^2$ so $V = \sqrt{\frac{2\mathrm{PE}}{C}} = \sqrt{\frac{600}{40 \times 10^{-6}}} = 3870\ \mathrm{V}$

(b) Power is energy over time: $P = \frac{E}{\Delta t} = \frac{300\mathrm{J}}{0.002} = 150\ \mathrm{kW}$

Electric current

Classes of conductors: Insulators & Conductors

Insulators

- Electrons not free to move
- High resistance
- Examples: plastic, wood, ultrapure water

Conductors

- Electrons free to move
- Low resistance: metals, impure water, humans, ionised gas
- Note: electrons are the charge carriers in metals, but in biological fluids, the charge carriers are often ions (both +ve and –ve)

Semiconductors

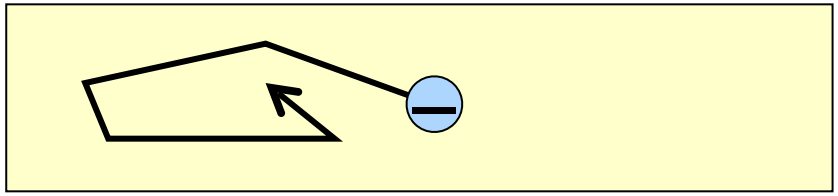
Superconductors

18 Electric Current

- Current = charge in motion
- Look at electric currents in detail…

<u>Isolated conductor</u>

- Random thermal motion of electron
- Speed $v \approx 10^6$ m/s

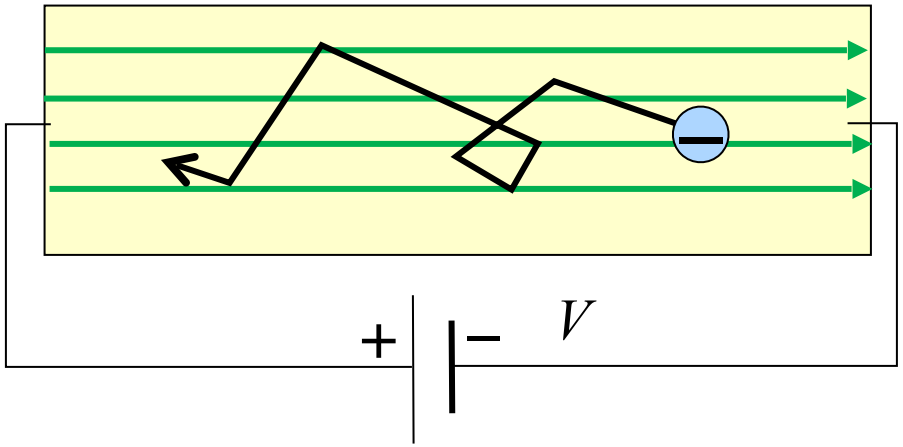


<u>Battery connected</u>

- Random thermal motion of electron unchanged
- *Drift speed* $v_d \approx 10^{-4}$ m/s superimposed

Internal electric field = force on charge = induced current

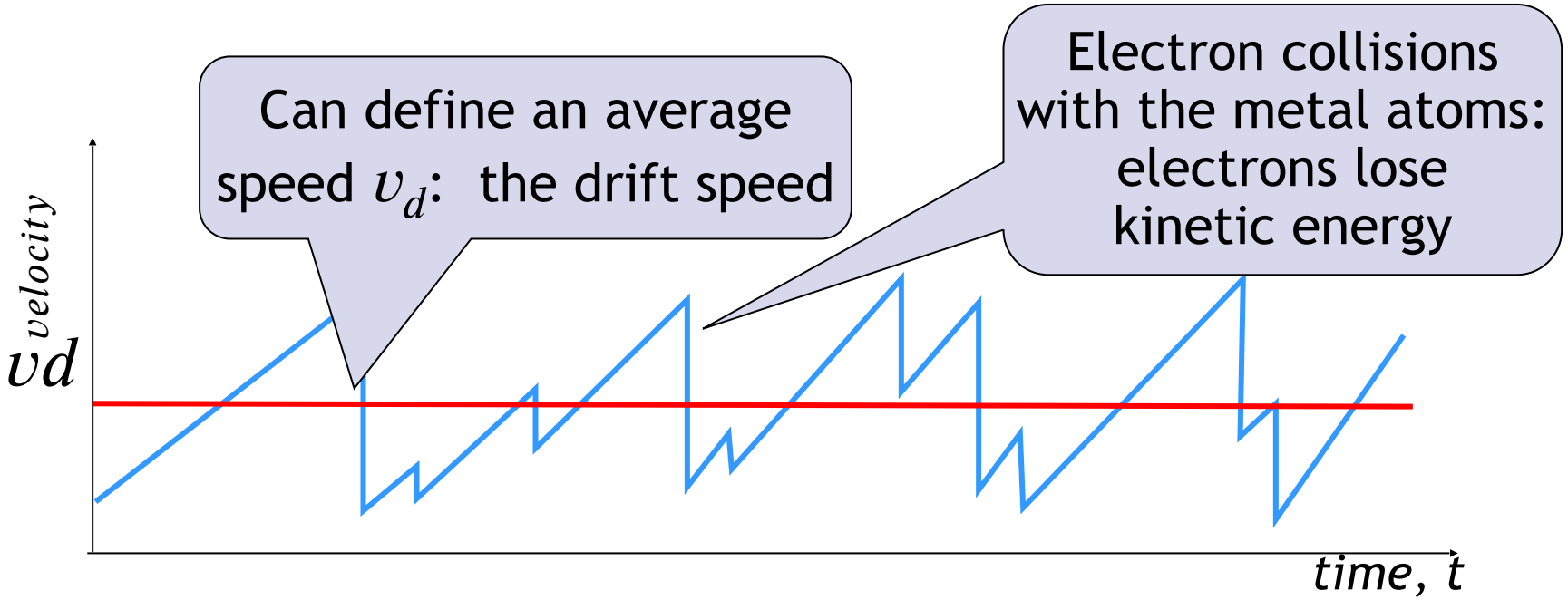
Internal field = potential difference



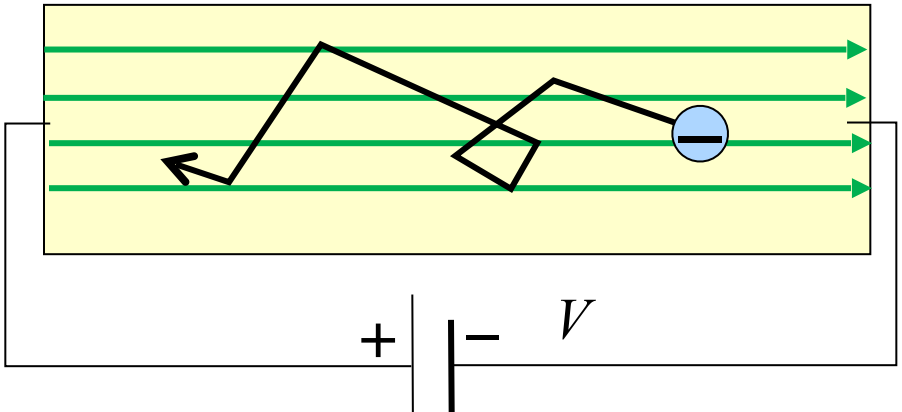
Demo Ed-2

Electric Current

- The electric field from the battery results in a force balanced by resistance, thus an average velocity



- How much charge is transported?



18.2 Definition of Current

- Define electric current as charge per unit time:

$$I = \frac{\Delta Q}{\Delta t}$$

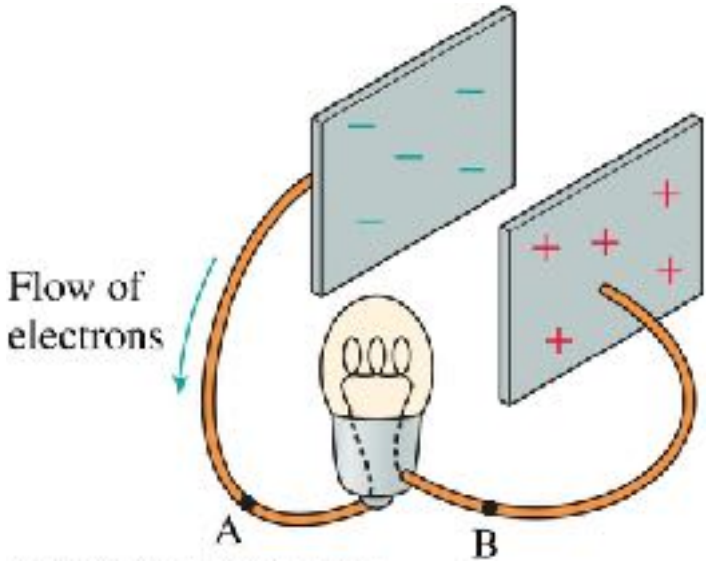
- Units: C/s or A (amperes, amps)

- 1 A = 1 C/s

- Electric current is analogous to water current
- Coulombs per second is like litres per second

Conservation of current

- Imagine a light bulb glowing as we discharge a capacitor



- The current leaving the light bulb is exactly the same as that entering it

Law of conservation of current:

– The current is the same at all points in a current-carrying wire

$$\sum I_{\text{in}} = \sum I_{\text{out}}$$

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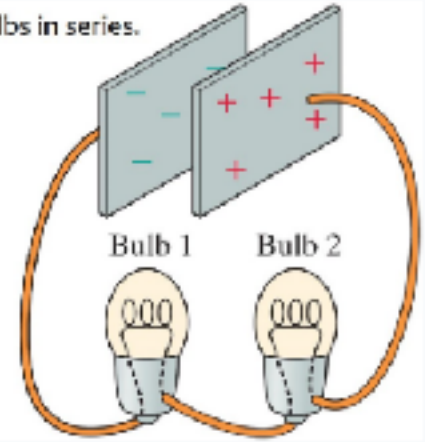
Bulbs in Series

Consider the circuit with two light bulbs in series.

Which of the following is true?



A. The current in bulb 1 is greater than the current in bulb 2

B. The current in bulb1 is less than that in bulb 2

C. The current is the same is each bulb

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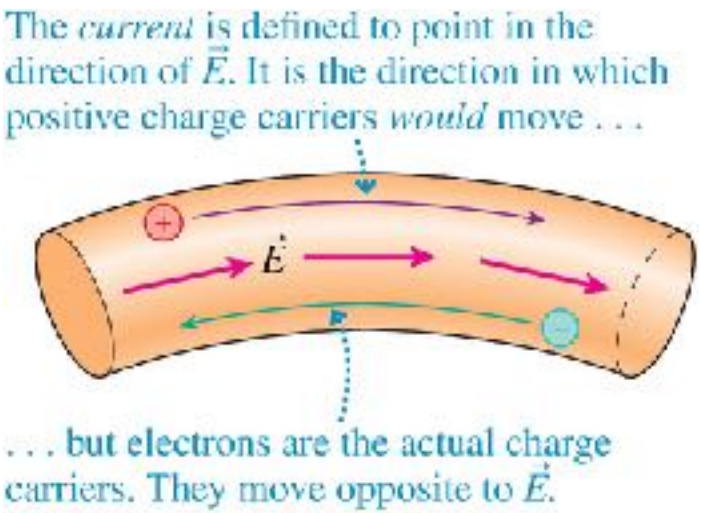
Definition of Current

- We define the current to be the flow of *positive* charge
- But…

- Usually it is electrons that move

- Blame Ben Franklin!

- Note that in fluids/electrolytes, charge carriers are both + and –



Example: charge flow in a light bulb

- A 240 volt 100 watt light bulb carries a current of 0.417 A. How much charge flows through the bulb in one minute?

$$Q = I\Delta t = 0.417 \times 60 = 25 \text{ C}$$

Summary

- Capacitors store charge $Q = CV$
- Parallel plate capacitors $C = \frac{\varepsilon_0 A}{d}$
- Dielectric materials reduce the magnitude of the electric field

  …and hence increase the capacitance $C = \kappa \frac{\varepsilon_0 A}{d}$
- The electrostatic potential energy of a capacitor is $\mathrm{PE} = \frac{1}{2} C V^2$
- Electric current is sustained motion of charge induced by an electric field
- Electric current is defined as $I = \frac{\Delta Q}{\Delta t}$

Summary....

- Batteries maintain a constant voltage difference emf = $\varepsilon = \Delta V$
- Most materials resist current flow
  - Resistance R depends on geometry and material $R = \frac{V}{I}$
- Voltage, current and resistance related through $V = IR$
- We assume connecting wires in circuits have zero resistance
- Power transformed in a resistor $P_R = IV$